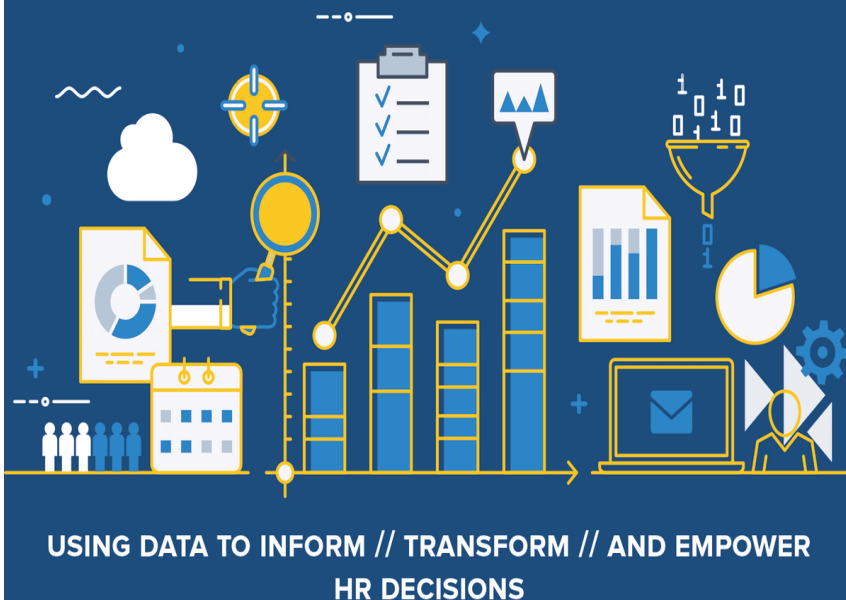


HR ANALYTICS



HUMAN RESOURCE PROBLEMS

ABSTRACT

The report is about the three contemporary business issues related to human resource management in any organization and the steps that can be taken to resolve those issues based on the data analysis, which can ultimately help to make important business decisions.

ASSESSMENT 1 – DATA PROBLEMS FOR BUSINESS CONTEXTS

TOPIC: DATA ANALYTICS – A KEY TO SOLVING CONTEMPORARY HUMAN RESOURCE OPERATIONAL PROBLEMS.

1. INTRODUCTION

The fundamental tenet of business studies is that management is a science and that data should be systematically and thoroughly investigated to guide management decisions (Martin & Golsby-Smith, 2017). According to Pinochet et al. (2021), as scholars and professionals seek to comprehend how data may be translated into valuable insights that improve organizational productivity, the notion and use of data and analytics in management have drawn more and more attention. Businesses are now more data-driven than ever because the digital revolution has generated massive data streams, giving them an unrivaled opportunity to reinvent their decision-making process (Soriano et al., 2018).

As per Exeed College (2019), Business Analytics can be defined as utilizing statistical approaches and data visualization tools for data analysis to get in-depth about various business operations to enhance business performance and productivity by making informed decisions. Analyzing the data through different sources and customer perspectives can help businesses achieve a competitive edge. Concerning this context, this report will depict the possible use of various data analysis tools and techniques to resolve contemporary business issues. Human Resource problems will be critically analyzed with the help of legitimate literature and applying data-driven management decisions to solve those issues.

The introduction includes a brief about business analytics, its relation to various operations, and its importance in informed decision-making. The second paragraph briefly discusses data analytics and its implementation into human resource operations. Further, section three shows the three human resource-relevant business problems, namely Employee Attrition, Employee Health & Safety, and Employee Diversity, that raise the necessity of data analytical thinking to help resolve these three problems and, in turn, can lead to making essential business decisions based on that data. Lastly, section four includes a conclusion and recommendations for the successful running of businesses.

ASSESSMENT 1 – DATA PROBLEMS FOR BUSINESS CONTEXTS

2. DATA ANALYTICS IMPLEMENTATION IN HUMAN RESOURCE MANAGEMENT

According to Barzilay (2018), in a business context, Data Analytics is a process of collecting, preparing, interpreting, and analyzing data to enable businesses to optimize their operations by looking for trends and making judgements about the future. In recruiting and hiring, data analytics has recently become a statistical study of figures that explains how to manage human capital and evaluate HR attempts to increase workforce productivity through training, development, and lowering employee turnover (Deeba, 2020). The use and interest in data analytics have surpassed various management disciplines, including Human Resource Management, by implementing HR Analytics to enhance data-driven decision-making (Fernandez & Gallardo-Gallardo, 2020). In the contemporary context, HR Analytics can be defined in simple terms as the effect of the workforce on the implementation of any organizational strategies (McCartney et al., 2022). Whereas, Gayathri et al. (2022) explain HR Analytics as the process which is not just limited to scrutinizing and enhancing elements of human capital but also involves the execution of various analytical tools and techniques on people data to help achieve data-driven decision-making and ultimately leads to better performance.

Various data analytics techniques can be used to run the data, depending upon the nature of the given data set. These techniques are Descriptive Analysis, Diagnostic Analysis, Predictive Analysis & Prescriptive Analysis, each serving a different purpose (Pinochet et al., 2021).

- ✓ Descriptive Analysis - This technique is the most straightforward and basic one, which helps with the data display, organizing information, and examining the patterns of the data, aiding in getting a sense of the periodicity of specific data being analyzed.
- ✓ Diagnostic Analysis - This technique supports the creation of plans to produce the desired outcome or prevent surprises and helps analyze the influence and impact of every action.
- ✓ Predictive Analysis - This technique is used to anticipate the outcomes based on past or previous data or trends using statistical equations & mathematical models and predict future trends in business based on those results. Predictive Analysis can be done in two ways which are Forecasting & Regression Analysis.

ASSESSMENT 1 – DATA PROBLEMS FOR BUSINESS CONTEXTS

- ✓ Prescriptive Analysis - This technique suggests the best possible alternative measures that can be undertaken to prevent a particular incident. When company decision-makers, such as CEOs, analyze and forecast complicated circumstances, this method facilitates more effective decision-making based on diverse concepts.

3. HUMAN RESOURCE MANAGEMENT PROBLEMS IN VARIOUS BUSINESS CONTEXTS

The following section will explain in detail the various human resource-related concerns where data analytics can be used to make important business decisions.

3.1 MANAGING EMPLOYEE ATTRITION PROBLEM IN THE HOSPITALITY SECTOR:

According to Bindu and Srikanth (2019), Attrition (Employee Turnover) is defined as the situation or event of employees quitting any organization, which is ideally measured as the “Attrition Rate.” The human workforce is the essential, priority, and ideal component of any business, which helps to achieve increased productivity and ultimately can help achieve sustained competitive advantage (Joy & Radhakrishnan, 2012). The employee attrition problem is found maximum in the hospitality sector for various reasons. Through various insights, significant reasons that were found to influence employee attrition include work pressures, lack of engagement, volatile job opportunities in the competitive world, company culture & work environment, HR policies, senior management, increased career mobility, long working hours with fewer salary packages, and rising employee aspirations (El-Rayes et al., 2020). All these will affect service quality and will not satisfy the clients and other employees, which can ultimately ruin the reputation of any firm (Bindu & Srikanth, 2019).

There is various tangible and intangible remuneration associated with employee attrition which includes recruitment cost, advertisement, interview cost, and training cost, along with the massive cost of loss of intellectual capital due to the loss of an employee to competitors (Joy & Radhakrishnan, 2012; Bennett et al., 1993; Bindu & Srikanth, 2019). So, efforts should be put forward by the organizations in order to retain the employees for a longer duration for enhanced performance.

ASSESSMENT 1 – DATA PROBLEMS FOR BUSINESS CONTEXTS

In order to find out the attrition rate of employees, various data variables can be taken into consideration, and their interrelations can be critically analyzed for decision-making by using analytical tools & techniques such as Chi-Square Analysis and Linear Regression Analysis through SPSS and ANOVA software (Joy & Radhakrishnan, 2012; Bindu & Srikanth, 2019). These variables can include demographics such as gender, age, education, marital status, ethnicity; job salary; job satisfaction, job position, size of the firm; promotion rate; the number of work hours, and work experience (El-Rayes et al., 2020; Joy & Radhakrishnan, 2012; Bennett et al., 1993). There are several solutions to this HR problem of reducing the attrition rate of employees. Firstly, monetary benefits such as pension plans, compensation, and vacation pay should be given to employees (El-Rayes et al., 2020). Secondly, as per Joy and Radhakrishnan (2012), an adequate workload should be laid per employees' capacity. Further, a positive work environment and transparency should be maintained, proper training should be provided, and timely appraisals based on performance should be carried out. Also, managers should provide specific financial benefits such as bonuses or overtime pay to the employees who work for more business hours and have low salaries (Bindu & Srikanth, 2019).

3.2 MANAGING EMPLOYEE HEALTH & WELLNESS PROBLEMS IN I.T.

SECTOR:

Wellness is a state of equilibrium between physical, spiritual, psychological, professional, economic, and societal requirements (Qaisar et al., 2018; Ryu, 2016). An organization that is considered healthy has excellent organizational efficiency due to its framework, culture, and administrative procedures (Qaisar et al., 2018; Singh & Jha, 2022). Various factors affecting employees' mental and physical health include work stress at the workplace, long working hours, employees' work not being recognized, and unachievable targets pressure without proper resources (Shava et al., 2021; Ryu, 2016). Poor health of employees can lead to a loss in productivity leading to bad organizational performance with economic implications of a decrease in profits (Singh & Jha, 2022). The highest employee health issues occur in I.T. industries due to high-stress levels.

Data analytics can be run to determine the relationship between employee health & organizational health. Variables such as employee demographics such as age, gender & designation; the number of working hours; job satisfaction rate; employee burnout rate; job salary; and employee productivity rate in terms of time management & happiness can be

ASSESSMENT 1 – DATA PROBLEMS FOR BUSINESS CONTEXTS

measured and critically evaluated by running factor analysis like EFA & CFA using either SPSS or Analysis of Moment Structure software (Qaisar et al., 2018; Shava et al., 2021; Ryu, 2016). Some of the solutions that can be recommended through data analytics for maintaining the good health and well-being of the employees are as follows. The managers should conduct wellness & health promotion programs to build awareness, evaluate health-based outcomes, and brief the employees regarding the firm's commitment towards them (Qaisar et al., 2018). Also, organizations should build a positive work environment by boosting their employees' confidence and encouraging them all the time (Singh & Jha, 2022). Further, benefits of work resources in terms of financial aid such as health insurance, bonuses, and retirement allowances should be given to the employees (Shava et al., 2021). Also, proper resources should be arranged for the employees, which can ease them to achieve their unachievable targets. Lastly, managers should provide psychological support for employees working longer hours and having fewer salary packages to reduce stress levels (Ryu, 2016).

3.3 MANAGING EMPLOYEE DIVERSITY PROBLEM IN THE EDUCATION

SECTOR:

Kundu et al. (2020) define Diversity as the distinction between people on any trait that influences how individuals view one another. Gender, age, race, ethnicity, national origin, cultural background, and others are among the differentiating factors. The highest diversity is found in the education sector, wherein people from various ethnic backgrounds, genders, and education work. Any company prioritizing gender equality has a better chance of retaining brilliant employees and improving performance (Zhuwao et al., 2019). Successful management of diversity requires employers to celebrate, value, and continually encourage diversity amongst employees rather than accepting simply as a concept. For senior managers of companies worldwide, managing diversity has thus become a key concern in generating sustained competitive advantage (Kundu & Mor, 2017). Not managing diversity issues results in sexual harassment, employee discrimination, and personal biases among colleagues, which can lead to reduced performance (Jaiswal & Dyaram, 2018; Zhuwao et al., 2019).

Data analytic techniques such as Descriptive Analysis and Regression Analysis can be used to measure the variables of diversity such as gender, ethnicity, mother tongue, age, religion, and firm's performance through ANOVA and SPSS Amos 22 software (Kundu et al., 2020; Kundu & Mor, 2017; Jaiswal & Dyaram, 2019). With the help of data analytics, the probable solutions

ASSESSMENT 1 – DATA PROBLEMS FOR BUSINESS CONTEXTS

which can be recommended to the managers of the organizations are that effective diversity management initiatives, including multicultural counseling and learning programs, should be put in place to reduce any detrimental effects that arise among the employees regarding in-group and out-group mentalities (Zhuwao et al., 2019). Also, in order to suit various ethnic communities inside the education sector, organizations should advocate the use of English as the medium of communication as a symbol of equality, due to which disputes over communication and ethnicity are avoided, which helps employees perform better (Kundu et al., 2020; Kundu & Mor, 2017).

4. CONCLUSION

To summarize the report, data analytics is essential for all business organizations to resolve the contemporary issues related to Human Resource Management (Pinochet et al., 2021). Human resource managers must use data analytics to boost the effectiveness of different HR procedures and initiatives, positively impacting the organization's business outcomes (Gayathri et al., 2022). Such techniques can help managers in the recruitment & selection process, workforce planning & developing culture in the organization as well as scheduling the training sessions and, in turn, can help in addressing the HR problems such as employee attrition, employee health & wellness & employee diversity (Barzilay, 2018; Cho et al., 2023). The benefits of using HR Analytics are that it helps in improving talent acquisition of the best candidates, develops policies to makes creates the best working environment for employees, thereby reducing the attrition rate, and helps in identifying the skill gaps among employees, thereby improving the productivity rate (*People Data Labs*, 2022). However, data analytics requires expert knowledge and equipment to successfully gather, handle, and evaluate data. The challenges faced while implementing HR Analytics are that getting consolidated people data from different departments of huge organizations gets tough. Also, security & compliance issues are there since the data of the employees is highly confidential. Lastly, since HR analytics is an intensive process, it requires specialized and trained IT professionals, which also incurs financial resources (*People Data Labs*, 2022).



Figure 1: *People Data Labs* (2022)

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